

Psychiatry CMS-Form 1

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3. A 37-year-old man comes to the physician for a follow-up examination 3 months after being diagnosed with alcoholic cirrhosis. He has missed three appointments that he attributes to transportation problems. His driver's license was suspended 3 months ago for driving under the influence of alcohol. He does not believe that his drinking is a problem, but he has cut down the amount he drinks from 12 beers daily to 4 beers daily. He is disheveled and restless and claims that he has not slept well for the past 3 months. His pulse is 96/min, respirations are 16/min, and blood pressure is 170/90 mm Hg. Physical examination shows no other abnormalities except for peripheral edema of the lower extremities. On mental status examination, he describes his mood as fine, but he is irritable and has poor eye contact, especially when discussing his drinking. He speaks at a normal rate and with normal response latency. His thought process is logical and coherent without evidence of psychosis. His serum albumin concentration is 3 g/dL, total serum bilirubin concentration is 2.6 mg/dL, and serum γ -glutamyltransferase (GGT) activity is 158 U/L (N=5–50). Which of the following is the most appropriate next step in management?
- A) Recommend alcohol rehabilitation
 - B) Clonidine therapy
 - C) Diazepam therapy
 - D) Disulfiram therapy
 - E) Furosemide therapy
 - F) Liver transplantation evaluation

4. A 47-year-old woman is brought to the emergency department by her husband because of increasing confusion during the past 2 days. On arrival, she has a generalized tonic-clonic seizure lasting 4 minutes. She has bipolar disorder treated with several medications, but her husband is unsure of their names. He knows that she sometimes takes ibuprofen for mild arthritis pain caused by exercise. He says that she has been active and exercising more lately, but he cannot think of other changes in her routine. She has no history of seizure disorder. She is oriented to person but not to place and time. Her pulse is 90/min, and blood pressure is 140/90 mm Hg. **On physical examination, she is tremulous and somnolent. There is bilateral nystagmus. An ECG shows a second-degree atrioventricular block.** The most likely cause of this patient's symptoms is an adverse effect of which of the following medications?

- A) Bupropion
- B) Haloperidol
- C) Lithium
- D) Risperidone
- E) Valproic acid

5. An 82-year-old man comes to the physician with his wife because of difficulty sleeping during the past year. He goes to bed around 10 pm and is able to fall asleep easily but awakens around 4 am and is unable to return to sleep. His wife says she has not noticed any changes in his sleeping habits, such as restlessness, snoring, or any other irregularity in breathing. He sometimes dozes off when he is reading a newspaper during the day or watching television at night. He has allergic rhinitis treated with loratadine as needed. His only other medication is a daily multivitamin. Vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, he is alert. He has a euthymic mood and reactive affect. He is pleasant and cooperative but appears anxious when describing his problem sleeping. His Mini-Mental State Examination score is 28/30. He says the day is Tuesday, but it is Wednesday, and he makes one error when performing serial sevens. In addition to providing instructions about sleep hygiene, which of the following is the most appropriate next step in management?

- A) Reassurance
- B) Neuropsychological testing
- C) Pulse oximetry
- D) Modafinil therapy
- E) Continuous positive airway pressure therapy
- F) Sleep EEG

6. A 32-year-old man is brought to the physician by his wife because of increasing confusion and agitation since he came home from work 2 hours ago. He first said he had a headache and then began pacing around the house, saying that he felt anxious. One week ago, he strained his back and began taking tramadol. He has a 4-year history of major depressive disorder treated with fluoxetine. He is anxious and diaphoretic. He is oriented to person but not to place and time. His temperature is 40°C (104°F), pulse is 110/min, respirations are 18/min, and blood pressure is 155/95 mm Hg. Physical examination shows dilated pupils, hyperreflexia, and generalized muscle rigidity. Dysregulation of which of the following neurotransmitters is the most likely cause of this patient's symptoms?

- A) Acetylcholine
- B) γ -Aminobutyric acid
- C) Dopamine
- D) Glutamic acid
- E) Norepinephrine
- F) Serotonin

7. A 13-year-old girl is brought to the physician by her father for a well-child examination. He is concerned about her weight and eating habits. He says that she is always "on the go" and never sits down for a full meal. She will often just grab a piece of fruit when she is on her way to an activity. She will eat a full meal when the family has dinner together every Sunday. She has talked about wanting to become a vegetarian because of her concern for animals. She has had the same group of friends since elementary school. She is the captain of her soccer team and practices 4 days weekly. She is active in an after-school drama program 3 days weekly and is the lead in this season's play. She maintains a B grade average. Her father says that she talks on the telephone "constantly" and is animated and cheerful most of the time. When she is in her bedroom, she prefers to keep the door closed and stops talking if she is on the telephone and her father walks into the room. She has a disorganized bedroom and often falls fast asleep on a bed covered with piles of clothes. Menarche has not occurred. She appears thin and muscular. She is at the 50th percentile for height and 35th percentile for weight, which is unchanged from last year. Vital signs are within normal limits. Breast and pubic hair development are Tanner stage 2. Physical examination shows no abnormalities. When interviewed alone, she is animated. She shrugs and rolls her eyes when asked about her diet and weight. She thinks she is fine and does not understand why her father is so upset. Which of the following is the most appropriate next step?

- A) Reassure the father that this is normal development
- B) Recommend individual psychotherapy
- C) Recommend nutritional counseling
- D) Recommend that the father keep a log of what his daughter eats
- E) Schedule weekly examination and weighing

8. A 21-year-old college student comes to the physician because of a 3-year history of distressing thoughts that he will hurt other people. He counts objects to prevent these thoughts from happening. These thoughts have become more frequent over the past 2 months, and he can no longer concentrate on his studies or sleep at night. Physical examination shows no abnormalities. On mental status examination, he states that he has violent sexual thoughts that he cannot control. He is oriented to person, place, and time, and his memory is intact. He says that he is embarrassed talking about these thoughts. Which of the following is the most appropriate pharmacotherapy?

- A) Amitriptyline
- B) Clomipramine
- C) Diazepam
- D) Haloperidol
- E) Lithium carbonate

9. A 37-year-old woman is admitted to the hospital after a suicide attempt by an overdose of pills and alcohol. She has a 6-year history of major depressive disorder treated with various antidepressants. During this period, she has attempted suicide by hanging herself twice, jumping from a bridge, and taking several overdoses of pills. She is 163 cm (5 ft 4 in) tall and weighs 50 kg (110 lb); BMI is 19 kg/m². Her pulse is 64/min, and blood pressure is 110/70 mm Hg. Mental status examination shows a depressed mood. She says that she has had difficulty sleeping and that she feels hopeless and helpless. There is still evidence of suicidal ideation. Laboratory findings are within the reference range. Urine toxicology screening is negative. This patient is most likely to have which of the following abnormalities?

- A) Decreased concentration of 5-hydroxyindoleacetic acid in cerebrospinal fluid analysis
- B) Delayed REM sleep on nighttime polysomnography
- C) Enlarged lateral ventricles on CT scan of the head
- D) Increased sensitivity to lactate infusion
- E) Serology positive for human leukocyte antigen-DR2

10. A 42-year-old woman comes to the physician because of suicidal thoughts, depression, emotional lability, and poor concentration. She is unable to recall when her symptoms began. Her father and sister have a history of similar symptoms. She also has had intermittent abdominal pain. She does not drink alcohol or take medications because they exacerbate her gastrointestinal distress. Abdominal examination shows no abnormalities. Neurologic examination shows decreased joint position sense. Romberg sign is present. Mental status examination shows a depressed mood. There is evidence of suicidal ideation. Which of the following is the most likely cause of this patient's symptoms?

- A) Acute intermittent porphyria
- B) Lyme disease
- C) Multiple sclerosis
- D) Somatization disorder
- E) Systemic lupus erythematosus

11. A 20-year-old college student comes to the physician because of a 1-year history of anxiety, dry mouth, sweating, nausea, and palpitations. She first noticed her symptoms when she tried to join a sorority and became embarrassed when she was asked to speak in front of the group. She says that she is failing a public speaking class because of overwhelming anxiety when she stands in front of the class to speak. She now avoids situations where she might have to speak in public. She says that her anxiety is "stupid" and thought taking a public speaking class would help with her symptoms. Physical examination shows no abnormalities. Mental status examination shows no abnormalities. Which of the following is the most likely diagnosis?

- A) Generalized anxiety disorder
- B) Obsessive-compulsive disorder
- C) Panic disorder
- D) Social phobia
- E) Specific phobia

12. A 20-year-old college student comes to student health services because she has been feeling "terrible" for the past week. She has been tearful and distracted since her boyfriend of 5 months broke up with her 1 week ago. She feels that she has a lot to offer in a relationship and is sad that things did not work out. She performed poorly on a test 3 days ago and has not been sleeping well. She asks if she should get something to help her sleep. She usually drinks four beers weekly, but she drank three beers 2 nights ago. She appears tired but well nourished and well groomed. Her pulse is 76/min, and blood pressure is 110/68 mm Hg. Physical examination shows no abnormalities. On mental status examination, she is tearful when discussing her relationship but becomes bright and engaged when she talks about her career plans and interest in art history. She thinks she is a good person and that hopefully things will get better. Her hematocrit is 34%, and mean corpuscular volume is 79 μm^3 . Urine toxicology screening is negative. Which of the following is the most likely diagnosis?

- A) Acute stress disorder
- B) Adjustment disorder
- C) Bereavement
- D) Borderline personality disorder
- E) Dysthymic disorder
- F) Major depressive disorder



13. A 57-year-old woman comes to the physician because of **progressive loss of voice** since her husband of 20 years died 10 months ago. She communicates with the physician in writing and reports that she initially assumed her hoarse voice was a result of her prolonged, loud crying, but the hoarseness has become increasingly more severe even though she no longer cries as much. During the past 2 months, her voice has been so weak that she has had to communicate by writing. Since her husband died, she has felt very sad and cries herself to sleep nightly. She has difficulty falling asleep initially but then sleeps through the night. She notes that it was her fault her husband died. On the day he died, they had argued because she had just found out he was having an affair. During their argument, she told him that she wished he would die. A few hours later, police informed her that he had died in a motor vehicle collision. Physical examination shows no abnormalities. On mental status examination, she has a blunted affect but becomes tearful and tense when addressing the argument with her husband. Which of the following is the most likely diagnosis?
- A) Adjustment disorder with depressed mood
 - B) Conversion disorder
 - C) Hypochondriasis
 - D) Major depressive disorder
 - E) Post-traumatic stress disorder

14. Fifteen minutes after administration of intramuscular haloperidol, a 21-year-old woman who is in the hospital has torticollis. She has a 1-month history of psychotic disorder and was admitted to the hospital 4 hours ago because of the acute onset of new delusions and agitated behavior. She appears frightened. Her temperature is 36.7°C (98°F), pulse is 100/min, respirations are 20/min, and blood pressure is 140/80 mm Hg. Physical examination shows lateral rotation of the head to the right with a right sternocleidomastoid muscle that is taut to palpation. Antagonism of which of the following neurotransmitters is most likely responsible for this patient's condition?

- A) Acetylcholine
- B) γ -Aminobutyric acid
- C) Dopamine
- D) Histamine
- E) Norepinephrine

15. A 57-year-old woman comes to the physician because of difficulty sleeping, tearfulness, and restlessness since her daughter was diagnosed with metastatic breast cancer 3 days ago. She reports that when she goes to bed at night, she is unable to fall asleep for several hours and lays in bed worrying about her daughter's situation. The patient underwent a mastectomy for breast cancer 7 years ago. She takes acetaminophen/butalbital for occasional migraines. Her vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, she is tearful and tense but calms during the conversation. There is no evidence of suicidal ideation. Which of the following is the most appropriate next step in management?

- A) Biofeedback
- B) Carbamazepine therapy
- C) Clonazepam therapy
- D) Clonidine therapy
- E) Imipramine therapy
- F) Olanzapine therapy
- G) Pentobarbital therapy
- H) Sertraline therapy



16. A 72-year-old woman with metastatic breast cancer is admitted to the hospital because of changes in her mental status during the past 3 days. Her family says she has been mumbling and softly moaning. They also report that she has been drinking more alcohol since she found out how far her disease has progressed. Current medications include fentanyl patches. There have not been any recent changes to her medication regimen. Vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, she appears to be hallucinating. She is not oriented to person, place, or time. Laboratory studies show:

Hematocrit	35%
Leukocyte count	8000/mm ³
Serum	
Na ⁺	137 mEq/L
Ca ²⁺	12.5 mg/dL
Urea nitrogen	15 mg/dL
Creatinine	1.1 mg/dL
Albumin	1.5 g/dL

Urinalysis shows no abnormalities. X-rays of the chest and a CT scan of the head show no abnormalities. Which of the following is the most likely cause of this patient's change in mental status?

- A) Adverse effect of fentanyl
- B) Alcohol withdrawal
- C) Hypercalcemia
- D) Major depressive disorder with psychotic features
- E) Meningitis

17. A 25-year-old woman is brought to the emergency department by police because she had been running in front of cars at a busy downtown intersection. She told police that she is on a mission to save the earth from pollution. On arrival, she is agitated and gestures animatedly as she swings her arms at staff in a threatening manner. She tells the staff that she needs to return to the street "to stop the internal combustion madness that's choking us all with smoke." Her medical history is unknown; she says she does not take any medications. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 140/90 mm Hg. Physical examination shows no abnormalities. On mental status examination, her speech is increasingly loud and fast. She reports that she hears the voice of Mother Earth asking for her protection. The most appropriate initial step in pharmacotherapy for this patient's acute symptoms is administration of which of the following?

- A) Fluoxetine
- B) Hydroxyzine
- C) Lamotrigine
- D) Lithium carbonate
- E) Ziprasidone

18. A 37-year-old woman is brought to the emergency department by police after local shopkeepers complained that she had been begging in front of their stores for 4 days. She appears malnourished, dirty, and confused. She is friendly and cooperative but is unable to give her name. She says that she is hungry and had been begging for food and money. Physical examination shows no abnormalities except for mild dehydration. On mental status examination, she is oriented to place and time but is unable to recall her own identity or where she lives. She has a full affect and a neutral mood. Psychomotor activity is normal. Her thoughts are goal directed. Her hematocrit is 45%, and serum sodium concentration is 145 mEq/L. An MRI of the brain shows no abnormalities. Which of the following is the most likely diagnosis?

- A) Complex partial seizures
- B) Dissociative fugue
- C) Malingering
- D) Schizophrenia
- E) Transient global amnesia

19. A 10-year-old boy is brought by his parents for a well-child examination. His parents report that he is rambunctious with a high energy level, and he has been retained in one grade. He sleeps approximately 6½ hours nightly. He has always been a handful and difficult to discipline; he argues about everything. For the past 2 years, he has often refused to go to school, so that his father then has to drive him there and walk him into his classroom. Several times, he has left school during the day. One month ago, he killed the family cat when he was playing too roughly. Although he said that it was an accident, he is not remorseful. His older brother is currently in a juvenile detention facility for grand larceny. The patient is at the 50th percentile for height and weight. His vital signs are within normal limits. Physical examination shows several ecchymoses in various stages of healing over the extremities. During the examination, he picks his nose. On mental status examination, he is initially insolent, but then he describes how he worries at night that something will happen to someone in his family. Which of the following is the most likely diagnosis?

- A) Adjustment disorder
- B) Antisocial personality disorder
- C) Attention-deficit/hyperactivity disorder
- D) Conduct disorder
- E) Generalized anxiety disorder
- F) Learning disorder

20. A 17-year-old girl is brought to the physician by her mother because her daughter has lost 4 kg (9 lb) during the past month and has seemed increasingly irritable and fatigued. Her mother reports that during the past 2 weeks her daughter also has been drinking more water and going to the bathroom more frequently. The patient's last menstrual period was 4 months ago. Menarche was at the age of 14 years, and her menses previously occurred at regular 28-day intervals. She says that she often feels cold and has occasional constipation and abdominal pain. She has no history of serious illness and takes no medications. She appears malnourished. She is 160 cm (5 ft 3 in) tall and weighs 41 kg (90 lb); BMI is 16 kg/m². Her temperature is 35.8°C (96.5°F), pulse is 55/min, respirations are 8/min, and blood pressure is 90/55 mm Hg. The skin is dry. Abdominal examination shows no guarding or rebound. On mental status examination, she has a depressed mood and an irritable affect. She says it upsets her when her mother suggests that she does not eat enough. She worries about being overweight and reports that she recently has been bingeing on large amounts of food and then purging through self-induced vomiting. Laboratory studies show:

Hemoglobin	12 g/dL
Hematocrit	35%
Mean corpuscular volume	90 μm^3
Leukocyte count	4000/mm ³
Serum	
Na ⁺	140 mEq/L
K ⁺	3.4 mEq/L
Cl ⁻	93 mEq/L
HCO ₃ ⁻	32 mEq/L
Mg ²⁺	1.4 mEq/L
Cholesterol	250 mg/dL
Thyroid-stimulating hormone	3.9 $\mu\text{U/mL}$
Thyroxine (T ₄)	6 $\mu\text{g/dL}$
Triiodothyronine (T ₃)	95 ng/dL

An ECG shows sinus bradycardia. Which of the following is the most likely diagnosis?

- A) Anorexia nervosa
- B) Bulimia nervosa
- C) Celiac disease
- D) Delusional disorder

21. A 16-year-old high school student will not eat meals with her family, preferring to eat alone in the kitchen. Her mother reports that she found food hidden under her daughter's bed. The patient does well in school, but her mother is concerned because her daughter does not date. The patient is 168 cm (5 ft 6 in) tall and weighs 44 kg (96 lb). She is most likely to have a history of which of the following?

- A) Amenorrhea
- B) Dental caries
- C) Diarrhea
- D) Hyperthermia
- E) Tachycardia

22. A 55-year-old woman is brought to the emergency department by her parents because of strange behavior for 3 months. The parents say that she has insisted that her father and the next-door neighbor have been taking money away from her. She also says that fumes that her neighbors are blowing under her door are giving her headaches and changing her skin tone. On numerous occasions, she has called radio stations and newspapers about these events. She insists that she is healthy and is acting normally. She has no previous history of psychiatric disorders. Mental status examination shows no incoherence, loose associations, or hallucinations. She is oriented to person, place, and time, and her memory is intact. Which of the following is the most likely diagnosis?

- A) Delusional disorder
- B) Major depressive disorder with psychotic features
- C) Paranoid personality disorder
- D) Schizophrenia, paranoid type
- E) Schizophreniform disorder

23. A 52-year-old woman comes to the physician with her husband because of a 6-month history of difficulty sleeping and fatigue. She describes an uncomfortable sensation in her calves that she first notices a few minutes after going to bed; the sensation can keep her awake for up to 2 hours. Although she occasionally snores enough to awaken her husband, once asleep, she sleeps soundly. She has had migraines since she was a teenager and currently takes sumatriptan approximately once monthly. Menopause occurred 6 years ago. She is 160 cm (5 ft 3 in) tall and weighs 59 kg (130 lb); BMI is 23 kg/m². Her pulse is 70/min, and blood pressure is 110/70 mm Hg. Physical examination shows a small patch of eczema over the right antecubital fossa. Mental status examination shows mild anxiety. She is concerned that her fatigue is affecting her work performance. Her husband recently began to consider retirement, and she knows how much he is counting on her salary. She gets discouraged when she cannot sleep and recalls crying one night when she became particularly frustrated. Results of laboratory studies are within the reference range. Which of the following is the most likely explanation for these findings?

- A) Conversion disorder
- B) Dysthymic disorder
- C) Generalized anxiety disorder
- D) Obstructive sleep apnea
- E) Restless legs syndrome
- F) Normal sleep phenomena

24. A healthy 4-year-old girl is brought to the physician by her father because of behavior problems during the past 6 months. He reports that his daughter has been having increasingly frequent temper tantrums and has shown increased disobedience and active and angry defiance. Last night, she threw her dinner plate across the table at her younger brother. He notes that her behavior is appropriate at day care and when she visits her grandmother. She has not been hyperactive. Physical examination shows no abnormalities. During the examination, the patient is generally compliant with the physician's directions. Which of the following is the most appropriate next step in management?

- A) Reassure the father that this is normal behavior
- B) Begin parent management training
- C) Begin lithium carbonate therapy
- D) Begin methylphenidate therapy
- E) Contact child protective services

25. A 46-year-old woman is brought to the emergency department in respiratory depression 30 minutes after attempting suicide by ingesting a large amount of an unknown drug. On arrival, she is intubated and mechanically ventilated. She is unconscious but responds to sternal rub by moaning. Examination shows pinpoint pupils. Diffuse crackles are heard on auscultation. Which of the following is the most likely drug ingested?

- A) Barbiturate
- B) Benzodiazepine
- C) Opioid
- D) Phenothiazine
- E) Tricyclic antidepressant

26. A 42-year-old man comes to the physician for a routine health maintenance examination. He has had fatigue, anxiety, and decreased work performance since his divorce 2 months ago. He has not gained or lost weight and has not been depressed. He takes no medications, and he does not smoke or use drugs; he drinks one or two alcoholic beverages weekly. He has difficulty falling asleep once or twice weekly. There is no family history of psychiatric illness. He appears anxious, but his affect is otherwise normal. Physical examination is noncontributory. Complete blood count, biochemical profile, and thyroid function tests are within normal limits. Which of the following is the most appropriate next step in management?

- A) Biofeedback
- B) Psychotherapy
- C) Administration of fluoxetine daily
- D) Administration of flurazepam nightly
- E) Administration of trazodone daily

27. A 22-year-old woman comes to the physician because of a 3-year history of intermittent headaches and diarrhea that have become more frequent and intense during the past month. She reports that after college graduation 3 months ago, her boyfriend of 3 years ended their relationship and left to take a job in another city. Two weeks ago, she started working in an entry-level accounting position for a local company. She is worried about how she will perform at her new job and whether she will have enough money to pay her bills and her school loans. She recalls that for much of her final year of college she was concerned about whether she would graduate and find a job. Her many thoughts often made it difficult to sleep and concentrate on her schoolwork. She states that she has always been a worrier and would frequently ask her boyfriend for reassurance that he would not leave her. She says that her friends describe her as often irritable and restless but bright. She has no history of serious illness and takes no medications. She does not drink alcohol or use illicit drugs. Her temperature is 36.9°C (98.4°F), pulse is 85/min, respirations are 10/min, and blood pressure is 135/85 mm Hg. Physical examination shows no abnormalities except for muscle tension in the neck and shoulder. On mental status examination, she appears restless and anxious. She says that her mood varies depending on what she is doing at the moment. Her thought process is organized, and there is no evidence of suicidal ideation or psychosis. Which of the following is the most likely diagnosis?

- A) Acute stress disorder
- B) Adjustment disorder
- C) Attention-deficit/hyperactivity disorder
- D) Dependent personality disorder
- E) Generalized anxiety disorder
- F) Major depressive disorder
- G) Somatization disorder

28. A 5-year-old girl is brought to the physician by her parents because of difficulty learning. They are concerned that she is not "catching on" as quickly as the other children in her kindergarten class. She is confused even by simple homework assignments and struggles to write the alphabet. Her teacher describes her as a happy, eager-to-please child who is having trouble learning. She appears to enjoy going to school and carries her backpack with pride. She wears glasses. She is at the 30th percentile for height and 65th percentile for weight. Vital signs are within normal limits. Physical examination shows epicanthal folds. A grade 2/6 murmur is heard best at the left sternal border. She can run well but climbs stairs by placing both feet on each step before proceeding to the next one. She has difficulty hopping on one foot. She can recite the alphabet and identify half of the letters. Which of the following is the most likely diagnosis?

- A) Angelman syndrome
- B) Down syndrome
- C) Fetal alcohol syndrome
- D) Fragile X syndrome
- E) Prader-Willi syndrome
- F) Rett disorder

29. An 8-year-old boy is brought to the physician by his mother because of a 2-week history of **disruptive behavior** in school. His teacher reports that he has been screaming out during class for no apparent reason. He has a 4-year history of attention-deficit/hyperactivity disorder. He takes no medications. Last year, he was seen by an allergist because of sniffing, clearing his throat, and rubbing his nose repeatedly. His symptoms resolved spontaneously within 1 month. Examination today shows intermittent, stereotyped, purposeless movements, including grimacing, raising his eyebrows, and shrugging his left shoulder. After 2 days of treatment with haloperidol, his symptoms improve. Which of the following is the most likely underlying cause of this patient's response to pharmacotherapy?

- A) Decreased binding of acetylcholine at the postsynaptic receptor
- B) Decreased half-life of acetylcholine
- C) Decreased release of acetylcholine from the presynaptic terminal
- D) Decreased binding of dopamine at the postsynaptic receptor
- E) Decreased half-life of dopamine
- F) Decreased release of dopamine from the presynaptic terminal

- G) Increased binding of acetylcholine at the postsynaptic receptor
- H) Increased half-life of acetylcholine
- I) Increased release of acetylcholine from the presynaptic terminal
- J) Increased binding of dopamine at the postsynaptic receptor
- K) Increased half-life of dopamine
- L) Increased release of dopamine from the presynaptic terminal

30. A 27-year-old teacher comes to the physician requesting a prescription for sleep medication. She has been unable to fall asleep until 2 or 3 am since the start of the school year 3 months ago. When she awakens at 6 am, she feels drowsy and remains tired throughout the morning. Last week, she fell asleep at her desk while her students were watching a movie. She believes that her job is in jeopardy. Drinking one glass of wine before going to bed has not helped her fall asleep. She recently took one of her mother's zolpidem tablets; she only slept for 2 hours and was then awake until 3 or 4 am. During vacations and on weekends, she typically sleeps from 3 am until noon and feels energetic and well rested. She currently does not take any medications or drink alcohol. Physical examination shows no abnormalities. On mental status examination, she appears tired and is teary. She can recall two of three objects after 5 minutes. Which of the following is the most likely diagnosis?

- A) Adjustment disorder with depressed mood
- B) Circadian rhythm sleep disorder
- C) Dysthymic disorder
- D) Major depressive disorder
- E) Narcolepsy
- F) Primary insomnia
- G) REM sleep behavior disorder

31. A 42-year-old woman comes to the physician because of a 2-week history of daily episodes of dizziness and feeling as though she were about to pass out. The episodes last for less than 5 minutes and are accompanied by a flushing sensation throughout her body. She says she is afraid to go to sleep because she fears that her heart will stop and she will never wake up. She suspects that her symptoms are caused by stress and may indicate recurrent heart problems. During the past 3 months, she has gone to the emergency department four times for treatment of chest pain. Evaluation each time showed no abnormalities, and she was told that she was taking too many nitroglycerin capsules without benefit. She had a myocardial infarction 6 years ago and underwent stent placement. She does not smoke, drink alcohol, or use illicit drugs. Her temperature is 37.2°C (98.9°F), pulse is 105/min, respirations are 18/min, and blood pressure is 145/88 mm Hg. There is no venous distention. Cardiac examination shows no abnormalities. On mental status examination, she appears anxious and says that her mood has been down. She reports that any twinge in her chest makes her worry that she is going to die. An ECG shows sinus tachycardia. Which of the following is the most likely cause of this patient's current symptoms?

- A) Acute myocardial infarction
- B) Conversion disorder
- C) Coronary artery disease
- D) Major depressive disorder
- E) Panic disorder
- F) Primary hypertension
- G) Somatization disorder

32. A 3½-year-old boy is brought to the physician by his parents because he has continued to wet his bed approximately two times weekly since being daytime toilet trained at the age of 2 years. They report that their two daughters were toilet trained by the time they were 2 years of age and have never wet their beds since that time. They have tried all of the available treatments including a bed-wetting alarm system and waking the child up each night at midnight to have him go to the bathroom. They believe that he is wetting the bed just to make them angry. Which of the following is the most appropriate intervention?

- A) Reassurance that this is normal behavior
- B) Play therapy
- C) Family therapy
- D) Imipramine therapy
- E) Sulfamethoxazole therapy

33. A 47-year-old woman with **schizophrenia, paranoid type**, comes to the physician for a follow-up examination. She has been treated with haloperidol for 15 years, and her symptoms have been well controlled. She has no history of serious medical illness. Physical examination shows no abnormalities. On mental status examination, she has a neutral mood and a blunted affect. She reports no auditory or visual hallucinations. She says she knows a number of patients with schizophrenia who have tardive dyskinesia and says that she would like to decrease her own risk for this problem. Which of the following is the most appropriate next step in pharmacotherapy?

- A) Decrease the dosage of haloperidol
- B) Switch from haloperidol to aripiprazole
- C) Switch from haloperidol to carbamazepine
- D) Add citalopram to the current regimen
- E) Add lorazepam to the current regimen
- F) Maintain the current regimen

34. A 21-year-old woman is brought to the emergency department by police after threatening her boyfriend with a meat cleaver. She picked up the cleaver after he pulled her by the hair when she did not bring him a drink. She says that she cannot stand the way he orders her around. Two years ago, she was diagnosed with major depressive disorder and successfully treated with an antidepressant. Her boyfriend flushed her pills down the toilet 2 months ago, and she never refilled her prescription. She reports now feeling the same way she did prior to starting the medication. She was fired from her job 2 weeks ago when her company declared bankruptcy. She has no family in the area. Physical examination shows multiple ecchymoses over the extremities in various stages of healing. Pelvic examination shows multiple vaginal abrasions. On mental status examination, she is tearful and has a labile affect. She avoids eye contact and repeatedly says, "I am so weak." When asked about suicidal ideation, she states that she would kill her boyfriend rather than kill herself. She denies delusions but says she sometimes hears her dead mother's voice telling her to defend herself. Which of the following is the most appropriate next step in management?

- A) Encourage the patient to press charges against her boyfriend
- B) Contact the patient's boyfriend to schedule couples counseling
- C) Recommend an outpatient support group
- D) Reinitiate antidepressant therapy
- E) Admit the patient to the psychiatric unit

35. A 67-year-old man comes to the physician because of a 2-month history of increased fatigue and decreased libido. He used to sleep 8 hours every night, but now he sleeps 10 hours every night and takes a nap during the day. He has had a 4.5-kg (10-lb) weight gain during this period. He has type 2 diabetes mellitus; hypertension; degenerative arthritis of the back, hips, and knees; and hypercholesterolemia. Current medications include metformin, ibuprofen, simvastatin, sildenafil, lisinopril, and hydrochlorothiazide. He is 178 cm (5 ft 10 in) tall and weighs 104 kg (230 lb); BMI is 33 kg/m². His pulse is 74/min, and blood pressure is 130/82 mm Hg. Pedal pulses are decreased. Examination shows Heberden nodes over the distal interphalangeal joints. Sensation to pinprick is decreased over the feet. Mental status examination shows a constricted affect. He says he is losing interest in life because his multiple medical problems are so difficult to manage. There is no evidence of suicidal ideation. Serum studies show:

Glucose	155 mg/dL
Cholesterol, total	162 mg/dL
HDL-cholesterol	46 mg/dL
LDL-cholesterol	90 mg/dL
Triglycerides	128 mg/dL

Which of the following is the most appropriate next step in pharmacotherapy?

- A) Add bupropion to the medication regimen
- B) Add insulin to the medication regimen
- C) Add venlafaxine to the medication regimen
- D) Discontinue lisinopril
- E) Discontinue simvastatin

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4 hr 17 min 51☐ Mark

36. A 13-year-old girl is brought to the physician by her mother because she has not yet had a menstrual period. Her mother states that the patient's older sister began menstruating at the age of 11 years. The patient has been treated for major depressive disorder with fluoxetine for the past 8 months. She is not sexually active. Physical examination shows no abnormalities. Sexual development is Tanner stage 2. Mental status examination shows mild depression and anxiety. Which of the following is the most appropriate next step in management?

- A) Discuss normal pubertal development
- B) Referral for reassessment by her psychiatrist
- C) Measurement of serum prolactin concentration
- D) Discontinue fluoxetine therapy
- E) Begin benzodiazepine therapy

37. A 47-year-old man is brought to the emergency department 1 hour after he inadvertently struck his head on a glass door that he mistakenly believed was open. He fell to the ground for a few seconds and said that he felt faint. His parents had to assist him to a chair. He did not lose consciousness and recalls the entire sequence of events. He was diagnosed with schizophrenia at the age of 17 years and was institutionalized until the age of 19 years. He has lived at home with his parents since that time. His only medication is **olanzapine**. His pulse is 80/min, and blood pressure is 120/70 mm Hg while supine; while standing, his pulse is 90/min, and blood pressure is 105/75 mm Hg. Physical examination shows a 2-cm circular ecchymosis on the left side of the forehead. Neurologic examination shows a fine tremor and occasional protrusions of the tongue. He is oriented to person, place, and time. A CT scan of the head is most likely to show which of the following?

- A) Accumulation of fluid in the subdural space in the left temporoparietal area
- B) Bilateral calcification of the nuclei basales
- C) Contusions of the frontal lobes
- D) Symmetric enlargement of the ventricles
- E) Tumor in the pituitary gland

38. A 27-year-old first-year surgical resident comes to the physician to request a prescription for a sleeping aid. He says that during the past 2 months, he has had increasing anxiety. On days when he is not on call, he spends hours reading medical texts because he cannot remember as well as he did in medical school. He avoids participating in social activities with friends. His wife has told him that he has been increasingly irritable. He falls asleep easily in the evening but wakes up multiple times during the night with the feeling that he has not read enough. He cannot sleep any later than 3 am even though he is tired. He and his wife now sleep in different bedrooms, so that she is not disturbed when he wakes up. He has had a 2.7-kg (6-lb) weight loss during this time. He recalls that he was always cheerful and reliable in medical school and did not experience this much anxiety. He has no history of serious illness and takes no medications. Physical examination shows no abnormalities. On mental status examination, he appears concerned and has a reactive affect. Which of the following is the most appropriate next step in management?

- A) Reassure the patient that this is a normal reaction to the first year of a residency program
- B) Advise the patient to consider alternatives to his surgical residency program
- C) Recommend over-the-counter diphenhydramine therapy as needed
- D) Begin escitalopram therapy
- E) Begin quetiapine therapy



39. A 20-year-old woman is brought to the emergency department immediately after being found comatose at her desk. During transport, a fingerstick blood glucose concentration is 15 mg/dL; after administration of 50% dextrose in water, she promptly becomes alert. She takes no medications. Her mother has type 2 diabetes mellitus. Examination shows no abnormalities. Her serum insulin concentration is increased, and serum C-peptide concentration is within normal limits. Serum concentrations of electrolytes, thyroxine (T_4), and cortisol are within the reference range. Which of the following is the most likely explanation for these findings?

- A) Insulinoma
- B) Reactive hypoglycemia due to fasting
- C) Surreptitious administration of insulin
- D) Surreptitious administration of an oral hypoglycemic agent
- E) Undiagnosed type 1 diabetes mellitus

40. A 47-year-old man is admitted to the hospital 20 minutes after he was involved in a motor vehicle collision. He was an unrestrained passenger. He has a history of cocaine abuse and meperidine dependence but has not used illicit drugs for 9 years. He takes no medications. Physical examination shows several deep lacerations on the lower extremities. There are no other injuries. Serum and urine toxicology screening is negative for opioids, cocaine, and alcohol. Extensive wound debridement and suturing are planned. Which of the following is the most appropriate next step in management of this patient's pain?

- A) Administration of gabapentin and amitriptyline
- B) Administration of ibuprofen
- C) Administration of morphine
- D) Acupuncture and application of cold packs
- E) Transcutaneous electrical nerve stimulation

41. A 42-year-old man is brought to the physician by his wife because of a 2-month history of staring spells that last 1 to 2 minutes each. During episodes, he also smacks his lips and picks at his shirt collar. Four years ago, he was comatose for 2 weeks after sustaining a head injury in a motorcycle collision; he required 6 months of rehabilitation. He reports that over the past year, he has had intermittent episodes of smelling burnt rubber; episodes occur approximately every 2 weeks. He also hears an intense hissing sound during these episodes. Examination shows no abnormalities. Which of the following is the most likely finding on EEG?

- A) Burst-suppression pattern
- B) Diffuse 3-Hz spike and slow wave activity
- C) Focal spikes localized to the temporal lobe
- D) Hypsarrhythmia
- E) Periodic lateralized epileptiform discharges
- F) Polyspike and slow wave activity
- G) Triphasic waves

42. A 23-year-old man with a 3-year history of **bipolar disorder well controlled with lithium carbonate** would like to discontinue drug therapy because of adverse effects. He is upset because he has gained 6.8 kg (15 lb) and developed acne. He has tried trials of other mood stabilizers that were ineffective in controlling his symptoms. He has a 2-year history of social isolation characterized by fears that he will be rejected by women because of his weight gain and acne. He recently graduated and is concerned about his ability to get a job because of mandatory drug testing. Which of the following is the most appropriate next step in management?
- A) Write a letter about lithium carbonate therapy for the patient to give to his employer
 - B) Explain the risk of illness recurrence without medication
 - C) Initiate cognitive behavior therapy
 - D) Switch to fluoxetine therapy
 - E) Switch to haloperidol decanoate therapy

43. A 7-year-old boy is brought to the physician by his mother because of difficulty at school since beginning first grade 6 weeks ago. His teachers state that he has been daydreaming frequently, does not complete his work, and does not participate in classroom discussions. His mother reports that he is very shy, will not go to sleepovers, and has stomachaches every weekday morning. He played on a soccer team, but he quit when his mother could not drive him to games and he had to carpool with another family. Physical examination shows no abnormalities. On mental status examination, the boy sits on his mother's lap, does not make eye contact with the physician, and is hesitant to answer questions but seems eager to please. Which of the following is the most likely explanation for these findings?

- A) Attention-deficit/hyperactivity disorder
- B) Dysthymic disorder
- C) Generalized anxiety disorder
- D) Separation anxiety disorder
- E) Social phobia
- F) Age-appropriate behavior

The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with hallucinations, select the most likely diagnosis.

- A) Amphetamine intoxication
- B) Amphetamine withdrawal
- C) Borderline personality disorder
- D) Hallucinogen intoxication
- E) Hallucinogen withdrawal
- F) Histrionic personality disorder
- G) Major depressive disorder with psychotic features
- H) Opioid intoxication
- I) Opioid withdrawal

44. A 27-year-old man comes to the emergency department after a motor vehicle collision. He states that he feels that he is being followed by the FBI. His temperature is 37.6°C (99.7°F), pulse is 96/min, respirations are 16/min, and blood pressure is 130/90 mm Hg. Physical examination shows mild hyperreflexia; there is no evidence of head trauma, and no other abnormalities are noted. On mental status examination, he is agitated with affective lability and rapid speech. He is oriented only to person and place; other tests of the mental status examination cannot be successfully completed.

For each patient with hallucinations, select the most likely diagnosis.

- A) Amphetamine intoxication
- B) Amphetamine withdrawal
- C) Borderline personality disorder
- D) Hallucinogen intoxication
- E) Hallucinogen withdrawal
- F) Histrionic personality disorder
- G) Major depressive disorder with psychotic features
- H) Opioid intoxication
- I) Opioid withdrawal

45. A previously healthy 25-year-old woman is brought to the emergency department by her girlfriend because of a sudden behavior change. She is anxious because "everyone has turned into spacemen." She avoids the medical staff, stating that they look like "large furry monsters." She sees "multicolored snakes singing Hail Mary" on the floor. Her temperature is 37°C (98.6°F), pulse is 76/min, respirations are 16/min, and blood pressure is 118/66 mm Hg. Mucous membranes are dry. Cranial nerves are intact. Neurologic examination shows normal findings except for mydriasis. On mental status examination, she is agitated, frightened, and fully oriented to person, place, and time. Toxicology screening is negative.

46. A 10-month-old boy is brought to the physician because of irritability throughout the night. His mother says that he has been crying and has refused to bear weight on his lower extremities since she picked him up from the babysitter the previous afternoon. She is concerned that he may have hurt himself while in the babysitter's care. Examination shows tenderness to palpation of the right thigh. An x-ray shows mild demineralization of bone and a spiral fracture of the right femur. Which of the following is the most likely mechanism of injury?

- A) Child abuse
- B) Ehlers-Danlos syndrome
- C) Osteogenesis imperfecta
- D) Osteopetrosis
- E) Rickets

47. A 32-year-old man comes to the physician because he thinks he might be losing his mind. For the past year, he has been having intrusive thoughts of killing his 4-year-old son. He does not want to do this and considers the thoughts ridiculous. He is becoming increasingly depressed and is worried that something is happening to him. He is a pastor at a local church and says that the thoughts are especially prevalent when he is reading scriptures and preparing for a sermon. He has not told anyone, including his wife, about "these repulsive ideas." He says that he is a good father and takes good care of his son. During his first year of college, he was treated briefly with counseling for an adjustment problem. He says that he does not think he has a family history of psychiatric illness, but his mother is considered to be overly anxious. Vital signs are within normal limits. Physical examination shows no abnormalities. Mental status examination shows an embarrassed, mildly anxious affect. Results of laboratory studies are within the reference range. Which of the following is the most likely diagnosis?

- A) Bipolar disorder
- B) Dysthymic disorder
- C) Generalized anxiety disorder
- D) Major depressive disorder
- E) Obsessive-compulsive disorder
- F) Schizoaffective disorder
- G) Schizophrenia

48. A 32-year-old man is brought to the emergency department by his roommate because of a 5-hour history of confusion and agitation. The patient says that bugs are crawling all over his skin. He has been using cocaine six times daily for the past 3 days. On arrival, he is hypervigilant and agitated. He is oriented to person, place, and time. His temperature is 38.3°C (101°F), pulse is 110/min, respirations are 32/min, and blood pressure is 150/105 mm Hg. Examination shows no other abnormalities. Urine toxicology screening is positive for benzoylecgonine. An ECG shows a prolonged QT interval. In addition to intravenous hydration and supportive measures, which of the following is the most appropriate pharmacotherapy?

- A) Amitriptyline
- B) Bupropion
- C) Buspirone
- D) Clonazepam
- E) Ziprasidone

49. A 24-year-old male college student comes to student health services for a follow-up examination 2 weeks after starting sertraline for treatment of major depressive disorder. Before starting the medication, the patient had a 6-month history of decreased motivation, lack of interest in academic work and fraternity activities, and feelings of sadness and worthlessness. He also had a decreased appetite and difficulty falling asleep at night. He had thought about suicide but made no attempts. Physical examination shows no abnormalities. On mental status examination, he says that he feels better but still has difficulty sleeping. Which of the following is the most appropriate next step in management?

- A) Maintain the current dosage of sertraline and schedule a follow-up examination in 4 months
- B) Maintain the current dosage of sertraline and schedule weekly follow-up examinations for the next month
- C) Maintain the current dosage of sertraline and add amitriptyline to the medication regimen
- D) Discontinue sertraline and begin fluoxetine
- E) Increase the dosage of sertraline and schedule a follow-up examination in 2 months

50. A 32-year-old woman comes to the physician for a follow-up examination 6 months after she was a passenger on an airplane that skidded off the runway into the harbor. Two people were killed; she escaped without serious injury. After the crash, she had insomnia, a depressed mood, nightmares about plane crashes, and daytime flashbacks that have been treated with clonazepam and paroxetine. Overall, her symptoms have improved, but she is still too anxious to fly. She works as a buyer for a large department store and was told by her manager that her job is in jeopardy because she is required to travel to Europe monthly. Physical examination shows no abnormalities. Mental status examination shows mild anxiety. Her thoughts are organized and goal directed. She has difficulty completing serial seven subtractions. In addition to continuing clonazepam and paroxetine therapy, which of the following is the most appropriate next step in management?

- A) Recommend that the patient switch to a job that does not require air travel
- B) Begin cognitive behavioral therapy
- C) Begin group therapy
- D) Begin psychodynamic psychotherapy
- E) Add quetiapine to the current regimen